

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

TUESDAY, 9 MAY 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) questions in Section B, marks will only be awarded for the FIRST questions in that section in the order the candidate submits the answers.

No calculator is allowed for this paper.

Answer questions only in the answer booklet provided.

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**SECTION A**

Q1. (a) The JavaFX and Event Handling questions.

- (i) State THREE (3) life cycle methods of JavaFX application. (3 marks)
- (ii) To create a GUI using JavaFX, which class must be extended and its corresponding package and which method in the Application class must be overridden? (3 marks)
- (iii) Rewrite the code fragments below by using **Lambda expression** to simplify the event handling for the button that set the fill of rec to GREEN when the button btn is clicked. (2 marks)

```
btn.setOnAction(new EventHandler<ActionEvent>() {
    @Override
    public void handle(ActionEvent event) {
        rec.setFill(Color.GREEN);
    }
});
```

- (b) Fill in the code below in the missing part to create a HBox to hold the four buttons and a VBox to hold the text area and the HBox. The expected output is shown in Figure 1 below. (4 marks)

```
public void start(Stage primaryStage) {
    Button vegies = new Button("Vegies");
    Button beans = new Button("Beans");
    Button rice = new Button("Rice");
    Button beer = new Button("Beer");
    TextArea text = new TextArea("This is a text area");

    /* Missing part
       Q1 (b) */

    Scene scene = new Scene(root, 300, 200);
    primaryStage.setScene(scene);
    primaryStage.setTitle("JavaFX");
    primaryStage.show();
}
```



Figure 1: Expected output

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (Continued)**

- (c) Create a simple JavaFX application that displays a rectangle and uses two buttons to **rotate** the rectangle. Every click will rotate the rectangle by **10 degrees** (clockwise or counterclockwise) as shown in **Figure 2**.

Complete the program by filling out the missing codes for parts: (i), (ii), (iii) and (iv). You may omit the **import** statement in your answer.

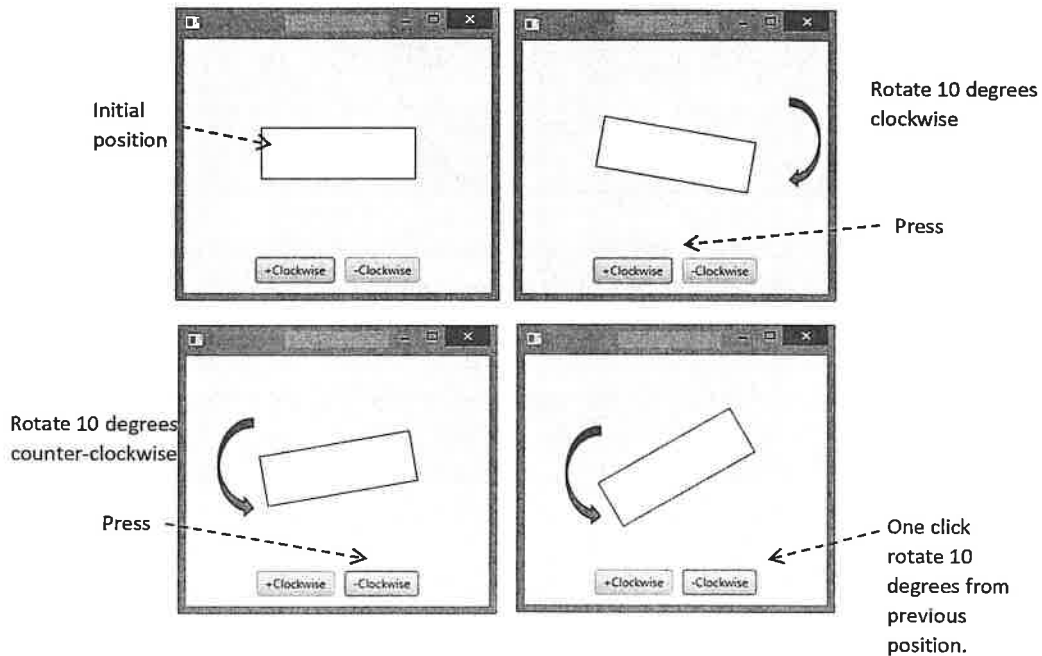


Figure 2

```
public class Rotate_Test extends Application {

    public void start(Stage primaryStage) {
        // Pane for placing the rectangle
        StackPane pane = new StackPane();

        // refer to (i)

        // HBox pane for holding the four buttons
        HBox hBox = new HBox();
        hBox.setSpacing(10);
        hBox.setAlignment(Pos.CENTER);

        Button btPos = new Button("+Clockwise");
        Button btNeg = new Button("-Clockwise");
        hBox.getChildren().addAll(btPos, btNeg);

        // refer to (ii)
    }
}
```

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (c) (Continued)**

```

        // Create a border pane and place pane and hBox in the
        border pane
        BorderPane borderPane = new BorderPane();
        borderPane.setCenter(pane);
        borderPane.setPadding(new Insets(10));
        borderPane.setBottom(hBox);
        BorderPane.setAlignment(hBox, Pos.CENTER);

        // refer to(iii)
    }

    // refer to (iv)
}

```

- (i) Create a Rectangle shape with size (150,50) and add it to the pane. Set the stroke of the rectangle to red and fill the rectangle with white color. (4 marks)
 - (ii) Add the event handling code for the two buttons using *Lambda expression*. Rotate the rectangle in the direction as indicated by the names of the buttons (see Figure 2) by 10 degrees when the button is pressed. (**Hint:** getRotate(), setRotate()) (6 marks)
 - (iii) Create a scene of size (300,250) to host the border pane, set an arbitrary title, place the scene into the stage and show the stage. (2 marks)
 - (iv) Add main() to launch the application. (1 mark)
- [Total : 25 marks]

Q2. (a) The following questions are based on File Input and Output,

- (i) What is the difference between **InputStream** and **OutputStream**? (2 marks)
- (ii) A text file "receipt.txt" stores item information of a purchase including item ID (String), item category (String), unit price (double), and number of units (int) as shown in Figure 3:

AT8546	Electronics	100.00	2
AT5432	Electronics	200.00	1
DB2298	Appliances	259.99	4
AT9912	Electronics	300.00	1
FC3876	Apparel	69.00	6

Figure 3: Text file of receipt.txt

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q2. (a) (Continued)**

Write a Java program to read from the text file, compute and output the **TOTAL price** (unit price * number of units) of all the **Electronics** items. For example, your program should output 700.00 ($100*2+200*1+300*1$) from the records as shown in Figure 3. You may omit the **import** statement in your answer.

(11 marks)

(b) The following question is based on Java Database Programming.

(i) What are the advantages of developing database applications using JDBC? (2 marks)

(ii) Given the following URL:
"jdbc:ucanaccess://Drive:/DatabaseDir/DatabaseLab.accdb" for a database, write a complete Java program (JDBC) to establish a connect to the database, create a statement, create a table named **Employee**, insert three records into the **Employee** table, query results, and display all the contents of the **Employee** table.

(10 marks)

// Expected Output:

Successfully connected to database
Table creation process successfully!

122	David	Engineer	Safety
132	John	Clerk	HR
143	James	Operator	QC

[Total : 25 marks]

Q3. Regarding to exception handling, answer the following questions.

(a) Briefly explain how to handle an exception in a program. Provide a simple coding example to illustrate your explanation. (3 marks)

(b) State the use of the keywords: throw and throws. (2 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q3. (Continued)**

- (c) Analyze the following code fragments and answer the following sub-questions:

```

public class Question3c {
    public static void main(String[] args) {
        try {
            String s = "50";
            int a = Integer.parseInt(s);
            int b = a/2;
            System.out.println("The value of b is " + b);
            System.out.println("Try block is finished");

        }
        catch (Exception ex) {
            System.out.println("NumberFormatException");
        }
        catch (RuntimeException ex) {
            System.out.println("NumberFormatException");
        }
        finally {
            System.out.println("The finally clause is
            executed");
        }
    }
}

```

- (i) Identify the error in the code. (1 mark)
- (ii) Fix the error by making a change to the code without adding or deleting any statements. Please explicit write down the change. (2 marks)
- (iii) What is the output displayed after the error has been fixed? (3 marks)
- (d) Thread and Multithreading,
- (i) How many ways to create a thread? Briefly state your answer. (3 marks)
- (ii) Given the code segment below, implement the necessary interface and write suitable codes to create an object of Demo class, create a Thread object and start the thread. (5 marks)

```

class Demo // implement the necessary interface here {
    public void run() {
        System.out.println("Thread is in Running
        state");
    }

    public static void main(String[] args) {
        // Missing codes
    }
}

```

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q3. (d) (Continued)**

- (iii) Analyze the following code fragments,

```
public class Count {
    int count;
    public void countMethod() {
        count++;
    }
}
```

If two threads try to access the value at the same time, will it cause conflicts? Briefly explain your answer and suggest if any changes needed. (3 marks)

- (iv) How do you create a thread pool with five fixed threads? How do you submit a task to a thread pool? (3 marks)

[Total : 25 marks]

SECTION B

This section consists of TWO (2) questions. Answer **ONLY ONE (1)** question.

- Q4. (a) Define the following terms with regards to *object-oriented programming*.

- (i) Class (2 marks)
- (ii) Object (2 marks)
- (iii) Constructor (2 marks)

- (b) Create a class named **Rectangle** to represent a rectangle. The class that contains:

- (i) Two **private** instance variables: *width* (double) and *height* (double). The default values are 1 for both width and height. (2 marks)
- (ii) A no-arg constructor that creates a default rectangle. (2 marks)
- (iii) A second constructor that creates a rectangle with the specified width and height. (2 marks)
- (iv) A public method named `getArea()` that returns the area of this rectangle. (2 marks)
- (v) A public method named `getPerimeter()` that returns the perimeter. (2 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q4. (Continued)**

- (c) Write a test program that creates two Rectangle objects. One rectangle with width 2 and height 20 and the other with width 5.5 and height 55.5. Display the area and perimeter of each rectangle as shown below. (6 marks)

//Expected output

The area of first rectangle is 40.0

The perimeter of first rectangle is 44.0

The area of second rectangle is 305.25

The perimeter of second rectangle is 122.0

- (d) Draw the UML class diagram for Rectangle class in 4(b). You should provide details of all the data fields and methods of the class (3 marks)
[Total : 25 marks]

- Q5. (a) A class has **static** variable and **instance** variable. Differentiate the two types of variables and describe how they are referenced differently in a class. (4 marks)

- (b) Create a **Vehicle** class that represents the type of vehicle, which contains the following requirements. Write your answer in a single Java class definition.

(i) A private integer data field named *numOfWheels*, representing the number of wheels. (1 mark)

(ii) A constructor that takes in one parameter that is the number of wheels. This class does not have a default constructor. (1 mark)

(iii) An accessor for the *numOfWheels* data field. (1 mark)

- (c) **Car** and **Bike** are two subclasses of the **Vehicle** class.

(i) The Car class has two private data fields: *numOfDoors* (int) and *color* (String), representing the number of doors and the color of the car. A constructor for creating car instance with the provided number of wheels, number of doors, and color of the car. Two accessors for the data fields. Write the class definitions for Car. (7 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q5. (c) (Continued)**

- (ii) The Bike class has a private `String` data field named `bikeType`. A constructor for creating bike instance with the provided number of wheels and type of bike. An accessor for the data field. Write the class definitions for Bike. (5 marks)
- (d) Draw the UML class diagram that shows the relationships of the above 3 classes in 5(b) and 5(c). Your class diagram should provide details of all the data fields and methods of each class. (6 marks)
- [Total : 25 marks]
-

